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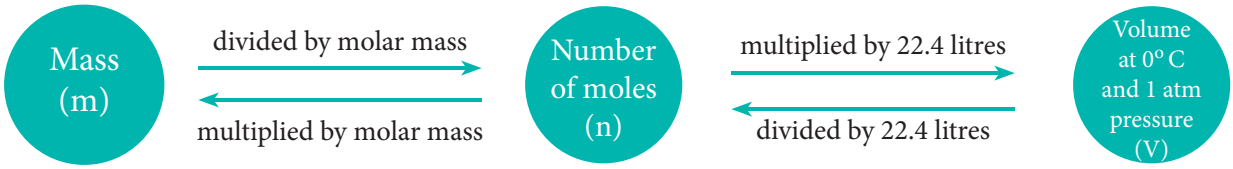
1.7.1 Stoichiometric Calculations:

Stoichiometry is the quantitative relationship between reactants and products in a balanced chemical equation in moles. The quantity of reactants and products can be expressed in moles or in terms of mass unit or as volume. These three units are inter convertible.



Let us explain these conversions by considering the combustion reaction of methane as an example. The balanced chemical equation is,

$CH_4\,(g) + 2\,O_2\,(g) \rightarrow CO_2\,(g) + 2\,H_2O(g)$

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Content	**Reactants**		**Products**	
	$CH_4\,(g)$	$O_2\,(g)$	$CO_2\,(g)$	$H_2O\,(g)$
Stoichiometric coefficients	1	2	1	2
Mole-mole relationship	1 mole	2 moles	1 mole	2 moles
Mass-mass relationship = no. of mole × molar mass	1 mol × 16 g mol^{-1} 16 g	2 mol × 32 g mol^{-1} 64 g	1 mol × 44 g mol^{-1} 44 g	2 mol × 18 g mol^{-1} 36 g
mass of reactants = mole × molar mass of reactants & Volume of product (1 mole of any gas occupy a volume of 22.4 litre at 273 K and 1 atm pressure)	16 g	64 g	22. 4 L	44.8 L
Volume - volume relationship	1 × 22.4 L 22.4 L	2 × 22.4 L 44.8 L	1 × 22.4 L 22. 4 L	2 × 22.4 L 44. 8 L